

Latent World Models for Language Reasoning

Action-conditioned prediction, counterfactual planning, and hierarchical control

TextJEPA project

July 14, 2026

Question and approach

Question. Can a reconstruction-free world model plan a language reasoning trace by predicting consequences in latent space?

State. Prompt plus outcomes established so far.

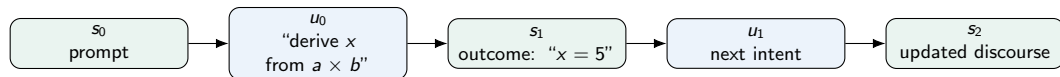
Action. A sentence describing the next reasoning intent, without its computed result.

Transition. Predict the latent state after that intent is executed.

Evaluation. At every step, enumerate the same feasible actions for all methods, score them, execute one, observe its rendered outcome, and replan.

The central methodological question is which training objectives are actually necessary for accurate action selection.

Reasoning as a controlled trajectory



The intent sentence contains operands and the operation, but not the answer. Consequently, predicting s_{t+1} requires modelling the consequence rather than copying outcome tokens.

A **distractor** is a feasible computation unrelated to the queried quantity. It may be locally valid while wasting planning budget.

Evaluation protocol

Candidate interface	Every policy scores all currently feasible intent sentences. This interface is identical for JEPA and language-model baselines.
Closed-loop planning	Execute one action, observe the true rendered outcome, re-encode the state, and replan.
Optimal action budget	Exactly n executions for a problem with n necessary computations. Any distractor is fatal.
+2-action budget	At most $n + 2$ executions. A successful trace can contain no more than two distractors.
Selection protocol	Choose objectives on validation data; evaluate the frozen model once on a disjoint generated test split.
Reporting	Three independent seeds for the final model, each final one-component ablation, and every principal baseline.

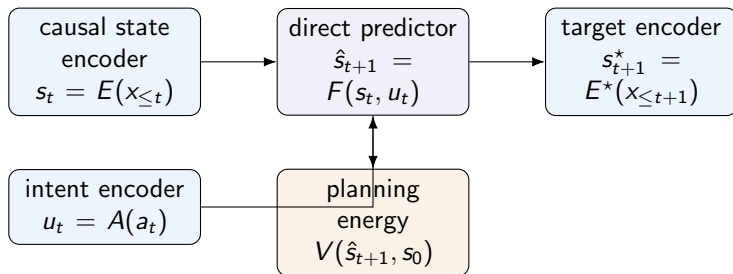
Matched policy baselines

Method	Decision rule
Random feasible policy	Uniform choice from the shared candidate set.
Autoregressive token policy	Mean token log-likelihood of each intent phrase, conditioned on the prompt and executed history.
Autoregressive sentence policy	Mean token cross-entropy of each intent.
Sentence policy + latent prediction	Intent likelihood with an auxiliary next-sentence latent target.
Latent dynamics model	Predict $F(s_t, u)$ and minimize its learned planning energy.
Symbolic preference reference	Same latent model, but action pairs are ordered by exact remaining computations; this is an annotated reference, not the proposed non-symbolic method.

Historical LMs that score fully rendered candidate outcomes are retained only as privileged diagnostics because those strings expose computed values.

Completed three-seed validation: token policy $0.827 \pm 0.003/0.978 \pm 0.003$; sentence likelihood $0.690 \pm 0.053/0.903 \pm 0.039$; adding latent prediction gives $0.817 \pm 0.034/0.953 \pm 0.010$ by likelihood and **0.840 ± 0.017** / 0.965 ± 0.013 by latent distance. The annotated reference reaches $0.962 \pm 0.013/0.997 \pm 0.003$. The reduced non-symbolic model reaches $0.797 \pm 0.008/0.963 \pm 0.008$ over three seeds. These are one-step, information-matched validation results; the final test remains sealed while the flat paper matrix is completed.

Intent-conditioned latent dynamics



In the over-complete model, replacing the residual parameterization by direct next-state prediction raises optimal-budget validation accuracy from .750 to .800. This result is not transported across objective sets: residual versus direct prediction must be reported as a matched ablation around whichever final objective set is selected.

Supervision taxonomy

Tier	Available information	Examples
Trace-only	Observed text and learned geometry	latent prediction, observed-action displacement decoding, terminal-distance comparisons
Environment interaction	Feasible actions and their rendered outcomes, but no quality annotation	counterfactual transition augmentation; geometric continuation teacher
Symbolic annotation	Task-specific relevance, remaining-step counts, or operation classes	exact preference ranking; scalar value regression; historical hybrid displacement labels

The selected objective set may use the first two tiers. Symbolic supervision is shown only as a reference and ablation boundary.

Multi-step latent-goal preference distillation

At one training state, compare the demonstrated root action with K sampled feasible alternatives. For each root u_i , construct an H -transition continuation using only environment interaction and target geometry:

$$q_{H,B}(u_i) = \min_{\pi \in \mathcal{B}_B(u_i)} d(E^*(x_{\leq t+H}^\pi), E^*(g_{\text{trace}})) .$$

At every continuation state, execute and target-encode every feasible child; $\mathcal{B}_B$ retains the B closest histories independently for each root. Train the planning energy to preserve the resulting pairwise order.

- ▶ H counts transitions including the root; $B = 1$ is greedy continuation.
- ▶ K counts alternatives, so training compares $K + 1$ root actions.
- ▶ The teacher never reads remaining-step counts or relevance labels.
- ▶ The goal is a demonstrated terminal state, so distractor-contaminated traces can make the proxy trajectory-specific.

Observed-action displacement decoding

The decoder reconstructs the complete observed action phrase from the latent displacement alone:

$$\hat{a}_t = D(s_{t+1} - s_t), \quad \mathcal{L}_{\text{LDAD}} = \text{CE}(\hat{a}_t, a_t).$$

It receives neither endpoint, the action embedding, nor an operation label. The head is removed at inference.

The long-horizon extension reconstructs an ordered action sequence:

$$D_H(s_{t+H} - s_t) \rightarrow (a_t, \dots, a_{t+H-1}).$$

This is the discrete-text analogue of raw-action LDAD; it is kept conceptually separate from inferred latent-action objectives.

Fixed-point objective selection

Start from a deliberately over-complete non-symbolic candidate and repeat:

1. Remove each remaining component individually.
2. Measure optimal-budget and +2-action validation accuracy.
3. Remove components whose worst loss is at most 0.02.
4. For effects in $(0.02, 0.05]$, compare a matched second-seed reference and ablation.
5. After replication, retain the component if its mean worst-case loss exceeds 0.02; otherwise remove it.
6. Combine accepted removals and begin another round around the smaller model.

Stop only when every remaining component has a reproducible measurable effect. No auxiliary—including terminal monotonicity, hierarchy, EMA, or open-loop prediction—has protected status.

First-round component elimination

Candidate modification	optimal	+2 actions	value τ
over-complete objective set	0.750	0.940	0.847
remove raw-action displacement decoding	0.755	0.960	0.874
remove open-loop latent prediction	0.840	0.970	0.920
remove geometry-to-value regression [†]	0.800	0.920	0.884
remove terminal monotonicity	0.780	0.950	0.894
remove on-trajectory outcome prediction	0.680	0.940	0.800
remove variance–covariance regularization	0.455	0.765	0.680
remove residual transition skip	0.800	0.960	0.891
remove EMA target network	0.575	0.840	0.735
remove macro-transition prediction	0.765	0.940	0.874
remove counterfactual transition prediction	0.755	0.935	0.858

Validation screen; table is seed 1. Seven removals were provisionally selected; outcome prediction, variance–covariance regularization, and EMA were retained. [†]At seed 2, removing geometry regression changes .785→.680 optimal-budget; removing faithful displacement decoding changes .785→.720. Matched reduced-model add-backs reject both terms: the reduced objective set is better without scalar geometry regression or raw-action reconstruction. Combining the provisional removals improves the reference from 0.750/0.940 to **0.805/0.970**. A second removal round retains each remaining term.

Remaining objectives have distinct functions

Intervention	optimal	+2 actions	transition match	value τ
none	.805	.970	.997	.904
latent-state prediction	.495	.825	.265	.671
latent-goal preference	.115	.500	.979	.080
outcome prediction [†]	.750	.942	1.000	.856
predicted-outcome rollout [†]	.772	.933	.998	.873
variance–covariance regularization	.615	.755	.953	.790
EMA target network	.720	.910	.987	.871
add geometry-to-value regression [†]	.750	.960	.999	.844
add raw-action reconstruction [†]	.788	.958	.999	.871

Latent-state prediction identifies action-conditioned transitions; the preference objective converts those transitions into a policy. Outcome targets leave transition identity intact but sharpen the selection energy and improve accuracy after an earlier distractor. Variance–covariance regularization remains load-bearing in the reduced model, affecting both transition organization and selection. EMA is also retained: its removal degrades both budgets and off-history recovery. Adding scalar geometry regression loses .048 optimal-budget accuracy over two seeds and remains absent. Faithful raw-action reconstruction also remains absent: its two-seed mean is .788/.958 versus .798/.968 without it. Both outcome terms are retained by matched two-seed comparisons. Removing the full target loses 0.048/0.025 under the optimal/+2 budgets; removing only its internal predicted-outcome rollout loses 0.025/0.035. [†]Two-seed means; the remaining rows are seed 1.

Terminal-distance monotonicity fails under distractors

The constraint asks every demonstrated transition to approach the particular trace-terminal embedding. A distractor may satisfy this geometric statement because that terminal state also contains the distractor outcome, while making no progress toward the query.

$H = 4$ teacher stratum	oracle top-1	pairwise τ_a
demonstrations without distractors	1.000	0.898
demonstrations with distractors	0.722	0.228

Removing the constraint currently outperforms retaining it with zero margin (0.780/0.950 versus 0.765/0.940). It is therefore excluded from the combined model and every subsequent removal round.

Longer greedy continuation is not automatically better

Teacher horizon	continuation after root	teacher top-1	optimal	value τ
$H = 1$	0	0.750	0.315	0.542
$H = 2$	1	0.930	0.750	0.847
$H = 4$	3	0.900	0.695	0.792
$H = 8$	7	0.830	0.620	0.728
$H = 16$	15	0.790	0.655	0.768

$H = 1$ shows that at least one environment continuation is necessary; this horizon is independent of hierarchy and open-loop prediction. Beyond $H = 2$, all width-one teachers remain worse. Potential causes are distinct:

- ▶ trajectory-specific terminal embeddings;
- ▶ error accumulation in geometry-greedy continuation;
- ▶ emitted preferences between actions tied under the task metric;

Holding the current reduced checkpoint fixed confirms $H = 2$ as the width-one optimum. Bounded continuation search changes the diagnosis: at $H = 4$, beam width $1/4/8$ raises teacher top-1 .810/.870/.900 and pairwise τ_a .646/.701/.728; the paired $H = 2, B = 1$ reference is .865/.661. At $H = 8/16, B = 8$, top-1 is .900/.895 and τ_a is .670/.672, so extending beyond four steps does not improve the teacher. Beam search remains non-symbolic and costs $O((K + 1)HBA)$ target encodings, where A is mean feasible branching. The matched $H = 2$ root sweep selects $K = 2$ alternatives; longer teachers are not promoted by this audit.

Margin tuning does not improve action selection

Intervention	optimal	+2 actions	oracle-order top-1	value τ
margin 0.5 (2-seed mean)	.798	.968	.946	.894
margin 0.25 (2-seed mean)	.763	.968	.941	.864
preference margin 1.0	.575	.870	.888	.781
preference margin 2.0	.435	.875	.820	.625
geometry filter .005	.740	.925	.943	.874
distractor mixture .30	.790	.935	.940	.882
scalar geometry regression [†]	.750	.960	.928	.844

The apparent seed-1 +2-action gain from margin .25 does not replicate: its two-seed optimal-budget mean is .035 lower with no +2-action gain. Margins 1/2 widen the median energy gap to .920/1.477 while reducing clean-history top-1 to .844/.749: they make the wrong order more confident. Retain margin .5. [†]Two-seed mean.

LDAD separates collapse prevention from transition identity

Target	regularizer	state std (off/on)	rank (off/on)	match (off/on)	RSA (off/on)
EMA	none	.457/.294	83.0/97.6	.432/. 959	.300/.737
online + stop-grad	none	.102/.343	2.9/101.8	.299/. 963	.406/.785
fully online	none	.000/.145	97.1/90.3	.322/. 988	.529/.778
fully online	VICReg	1.016/1.007	234.5/219.6	.334/. 980	.558/.909
fully online	SIGReg	.980/.975	94.2/103.9	.951/.998	.972/.956

VICReg can produce healthy scale and rank without identifying action-conditioned transitions. LDAD supplies transition identity; it does not by itself guarantee a calibrated planning energy. In the matched official symbolic-preference pilot, LDAD off/on raises transition matching .889→.956 but changes optimal-budget success .565→.535 and clean/after-error top-1 .848/.653→.837/.623.

Minimal two-objective text Delta-JEPA control

All rows use fully online gradients and unnormalized latent MSE, without EMA, stop-gradient, VICReg, SIGReg, outcome anchoring, hierarchy, or planning supervision.

Action objective	state std	state rank	transition match	RSA
none	0.00028	95.8	0.307	0.466
adjacent raw-action LDAD	0.117	89.5	0.975	0.775
four-action sequence LDAD	0.080	140.1	0.994	0.804

Raw-action displacement decoding prevents magnitude collapse and makes counterfactual transitions identifiable in the intended minimal setting.

Residual errors arise from action ordering

Both models identify counterfactual next states nearly perfectly. Symbolic preferences primarily improve their energy ordering, not transition identity.

Preference target	transition match	value τ	oracle-trace top-1	optimal success
latent-goal continuation	.997	.904	.948	.805
latent-goal continuation, gap .005	.997	.874	.943	.740
symbolic action order	1.000	.980	.995	.960

Preference target	clean-history top-1	after-error top-1	median margin
latent-goal continuation	.940	.706	.610
latent-goal continuation, gap .005	.914	.681	.520
symbolic action order	.989	.857	1.617

The mean problem requires 4.07 decisions and $0.948^{4.07} = 0.804$, matching the observed 0.805. Errors are rarely ties: only 1.2% of competitive decisions have an absolute margin below 0.02. Lowering the geometric tie filter to .005 degrades ordering and success, so the added small-distance comparisons are noisy rather than useful coverage. The remaining gap is a calibrated action-selection problem, with additional amplification after the first off-trajectory action.

Faithful iGSM transfer: separate model selection

The faithful domain uses the official generator and renderer rather than the stylized mod-23 environment. It changes sentence length, operation structure, graph depth, and action identifiers.

The transfer protocol repeats the complete scientific procedure:

1. initialize from the selected stylized model;
2. screen exact component additions: require a gain above .02, reject any budget loss above .02, and replicate gains in .02–.05;
3. repeat iterative one-component elimination to a fixed point;
4. compare matched autoregressive, random, symbolic-reference, and latent world-model policies over three seeds;
5. evaluate the frozen model on a disjoint official-iGSM test set.

Faithful iGSM: sentence likelihood is the strongest current baseline

One-seed validation policy	optimal	+2 actions
random feasible selection	.200	.460
token intent likelihood	.415	.675
sentence intent likelihood	.450	.755
sentence likelihood + latent prediction	.455	.765
same model, latent-distance selection	.445	.715
symbolic-preference latent reference	.565	.805

The auxiliary next-latent target changes likelihood selection by only .005/.010, and its latent distance is a worse selection score. Unlike the stylized domain, token-level likelihood also trails sentence likelihood. No official-iGSM configuration is frozen from these baseline rows; hierarchy is being resolved first in the faster stylized environment.

Variational studies are a separate question

Model		Action at training			Next-state target	Scientific question	
deterministic JEPA	discourse	observed	intent	sen- tence	deterministic latent	selected	planning model
probabilistic JEPA	discourse	observed	intent	sen- tence	diagonal Gaussian	uncertainty	with known actions
action-free stream	sentence	inferred posterior code			diagonal Gaussian	can modes	transition emerge without actions?

In the action-free model, $q(u_t | s_t, s_{t+1})$ observes the outcome. A learned prior $p(u_t | s_t)$ must recover a usable code before the outcome exists. Decoding the posterior code from $s_{t+1} - s_t$ is therefore not raw-action LDAD.

Observed-action variational LDAD does not prevent collapse

All cells below use a probabilistic next-state target, no VICReg/SIGReg, and the same observed intent action. Values show raw LDAD off/on.

Target gradient path	state std	shuffle ratio	LDAD-on token/exact	residual ²
online stop-gradient	.00075 / .00454	1.004 / 1.152	.898 / .582	1.006 / 1.012
fully online	.00034 / .00305	1.000 / 1.035	.900 / .583	1.004 / 1.007

VICReg here means variance–covariance stabilization; SIGReg is a random-projection Gaussianity regularizer.

The decoder recovers the observed sentence by amplifying tiny displacement directions, while learned variance calibrates a collapsed target mean. Consequently, faithful LDAD alone does not transfer the deterministic anti-collapse result to probabilistic states; EMA and explicit distributional regularization must be tested separately.

EMA and SIGReg expose action-conditioned uncertainty

All rows use probabilistic next-state prediction and pooled observed intents. Values show raw-action LDAD off/on.

Target / regularizer	state std	state rank	matched L1	shuffle ratio
EMA / none	.062/.286	76.4/52.3	.003/.008	1.182/1.455
EMA / VICReg	.844/.906	97.2/90.1	.016/.018	1.226/1.647
EMA / SIGReg	.977/.970	104.5/80.2	.418/. .062	1.375/ 2.293
official EMA / SIGReg	.981/.979	119.7/102.9	.463/. .067	1.105/ 1.462
online stop-gradient / SIGReg	.973/.946	88.5/45.1	.424/.008	1.398/1.219

EMA+SIGReg+LDAD gives healthy state scale and stronger observed-action dependence on both stylized and independently rendered official text. LDAD does not universally raise rank; action sensitivity, predictive error, and state geometry must all be reported.

A scalar schedule recalibrates long-horizon spread

The true intent sequence is fixed; 32 Gaussian trajectories are propagated recursively. One standard-deviation multiplier per horizon is fitted on 500 validation trajectories, then frozen and applied to 1,000 disjoint test trajectories.

Domain	LDAD	H	open-loop L1	raw 2σ	multiplier	calibrated 2σ
stylized	off	4	.533	.990	.742	.953
stylized	on	4	.144	.882	1.308	.954
official	off	8	.655	.998	.612	.953
official	on	8	.276	.776	1.684	.952

Raw-action reconstruction sharply improves conditional transition means. Its long-horizon under-dispersion is largely a scalar scale error: across all reported test cells, calibrated residual squared is .964–1.084 and two-sigma coverage is .943–.962. The scalar does not correct accumulated mean bias, so transition fidelity and uncertainty calibration remain distinct.

Same-state audit: broad support is not accurate control

For each state, encode every feasible rendered outcome. “Distinct” means a prediction enters the ball whose radius is half the distance to the nearest competing outcome; “precise” additionally requires posterior-level error.

Domain / target	posterior-code rec.	outcome sep.	posterior distinct	prior distinct	prior precise
stylized / stop-grad.+SIGReg	off	.706	.285	.052	.004
stylized / stop-grad.+SIGReg	on	.665	.645	.278	.016
stylized / zero free nats	off	.774	.006	.010	.010
stylized / four-component prior	off	.703	.230	.050	.005
official / EMA+VICReg	off	.005	.000	.000	.000
official / stop-grad.+SIGReg	off	.989	.753	.598	.073
official / stop-grad.+SIGReg	on	.743	.838	.675	.029

Posterior-code reconstruction separates outcome-informed modes and broadens support, but does not make it precise: on official iGSM it reduces precise coverage from .073 to .029 while improving broad coverage from .598 to .675. EMA+VICReg appears healthy globally while collapsing same-state alternatives; stop-gradient+SIGReg preserves alternatives but lacks a goal-directed prior. Forcing posterior–prior alignment by removing free nats collapses the learned action modes, so stronger KL is not pursued. A four-component Gaussian prior also collapses to 1.36 effective components and does not improve coverage; additional prior capacity cannot identify unseen same-state alternatives.

Architecture changes the probabilistic trade-off

All rows use EMA and the same 16-dimensional observed-action code. The ordered representation retains two dimensions per token; the direct predictor removes the residual state skip.

Reg.	action representation	predictor	LDAD	matched L1	shuffle ratio	state rank
SIG	pooled	residual	on	.062	2.293	80.2
SIG	ordered tokens	residual	on	.102	2.086	89.3
SIG	pooled	direct	off	.501	1.728	161.4
SIG	pooled	direct	on	.245	2.990	171.8
VIC	pooled	residual	on	.018	1.647	90.1
VIC	pooled	direct	off	.469	1.415	221.1
VIC	pooled	direct	on	.320	1.835	213.6

Ordered-token concatenation does not improve the selected setting. Direct prediction retains richer state geometry under both regularizers; residual prediction achieves the lowest error. SIGReg favors action dependence, while VICReg favors effective state rank. On official iGSM, direct-predictor LDAD reproduces the effect: matched L1 .719→.266 and shuffle ratio 1.097→1.457 at state rank 163.5. The ordered-token transfer is also negative: relative to pooled LDAD, exact phrase recovery improves .730→.775, but action-code rank halves 14.5→7.8, matched L1 worsens .067→.081, and the shuffle ratio falls 1.462→1.222. Official VICReg direct prediction confirms the trade-off: LDAD changes error .612→.199 and sensitivity 1.104→1.307 while retaining state rank 197.3. SIGReg yields stronger action dependence; VICReg yields lower error and richer geometry.

Direct prediction does not replace the target network

Fully online gradients with SIGReg; values show raw-action reconstruction off/on. A healthy representation requires scale, rank, and conditional action dependence simultaneously.

Predictor	state rank	action rank	matched L1	shuffle ratio
residual	14.1/17.5	11.6/11.5	.001/.002	1.054/1.138
direct	8.5/6.3	1.27/1.42	.004/.005	51.4/48.3

The large direct-predictor ratios are denominator effects: matched error is almost zero while action dependence occupies approximately one direction. Neither the direct architecture nor displacement-to-action reconstruction prevents low-rank collapse without a stabilized target network.

Hierarchical control protocol

Three observed intent actions are compressed into one macro action, and the high-level dynamics predict the corresponding state transition:

$$m_t = q_\phi(a_{t:t+2}), \quad \hat{z}_{t+3} = F_{\text{hi}}(z_t, m_t).$$

F_{hi} is a two-layer causal transformer that retains the complete macro-state/action history during autoregressive rollout. The primitive F_{low} is a Markov state-action MLP applied recursively; it does not attend to an explicit sequence of earlier primitive states.

1. **High-level planning.** CEM optimizes a sequence of macro actions, propagates them through F_{hi} , and scores the predicted macro states. Reported runs use 1,200 candidates, 20 refits, 10 elites, and variance EMA 0.9—within the hundreds-to-thousands/5–40-iteration HWM regime.
2. **Support control.** A conditional macro prior and a separately learned support score penalize implausible actions. Encoded terminal-state distance uses the solved trace as an oracle goal observation: it reproduces the HWM objective, but is not deployable when the answer is unknown. Exact graph distance is also labeled as oracle supervision wherever it trains a value.
3. **Closed-loop refinement.** The first predicted macro state becomes a subgoal for $Q_{\text{low}}(z, a, z_{\text{subgoal}})$, which scores only actions feasible *now*. One primitive action is executed and both levels replan. Near termination, a learned remaining-cost threshold switches to flat control.

No future feasible-action sequence is supplied to the deployed planner.

Dense supervision improves recursive dynamics

The recursive shifted-sequence loss supervises every valid origin at every horizon, rather than only a rollout from the initial state.

High-level LN-L1 error			
supervised depth N	$h = 1$	$h = 2$	$h = 3$
none	.586	.629	.684
1	.557	.600	.660
2	.559	.580	.631
3	.560	.578	.622

Low-level LN-L1 error			
supervised depth N	$h = 1$	$h = 4$	$h = 8$
none	.493	.637	.867
1	.470	.590	.707
4	.477	.552	.614
8	.480	.553	.604

On the same 200 long-tail problems, flat success changes from .300 to .335/.355/.345 for $N = 1/4/8$. Thus $N = 4$ is the behavioral choice even though $N = 8$ has the lowest eight-step prediction error. Better prediction is necessary but is not itself evidence that hierarchical control works.

State-value distillation is better calibrated than latent distance

All rows rank 11,659 valid macro spans at 200 held-out states. Exact remaining computations supervise the learned-value rows, so this is an *oracle-supervised diagnostic*, not the non-symbolic headline model.

High-level score	correlation	pair accuracy	top-1 optimal	regret
terminal latent distance (oracle goal observation)	.478	.591	.590	.545
macro action value $Q(z, m)$	.727	.694	.715	.405
state value $V(z')$, factual regression	.857	.729	.790	.295
$V(z')$ + counterfactual regression	.860	.734	.800	.280
$V(z')$ + regression + pairwise rank	.859	.735	.800	.280

The next-state value is substantially better calibrated than latent goal distance. Counterfactual regression helps slightly; an additional ranking hinge does not improve top-1 selection. A support classifier separates valid from three-standard-deviation perturbations almost perfectly, yet this offline separation does not by itself improve control.

A validation depth curve is not a hierarchy result

Dense high-level supervision and the recalibrated state value produce the first monotone planning-depth curve, but only on the 100-problem selection screen.

High-level score	one macro step	two macro steps	three macro steps
learned state value $V(z')$	.260	.300	.330
terminal latent distance (oracle goal observation)	.310	.310	.310
matched flat controller	.280	.280	.280

The learned value benefits from deeper recursive prediction; the HWM-style latent terminal distance does not.

This table is a model-selection diagnostic: one dataset, one checkpoint lineage, and only 100 episodes. It motivates a larger paired confirmation; it cannot establish a positive hierarchy effect.

Continuous latent-action hierarchy does not confirm

The selected setting uses low-level dense depth 4, high-level dense depth 3, the counterfactually trained state value, a soft support penalty, and three-step macro planning. Each row evaluates the flat and hierarchical controllers on the same 500 newly generated 7–9-step problems.

data / planner seed	flat	hierarchy	difference	macro decisions
31 / 731	.314	.272	-.042	.633
32 / 732	.336	.320	-.016	.623
33 / 733	.322	.274	-.048	.634
mean	.324	.289	-.035	.630

The 100-problem .33 versus .28 comparison was selection noise. Continuous latent-vector optimization followed by low-level subgoal refinement is 3.5 points worse on average and increases distractor selection. This rejects that particular HWM-style interface; it does not reject hierarchy in general.

What the hierarchy diagnostics rule out

Hypothesis	Decisive observation	Status
CEM is under-budgeted	1,200 candidates $\times$ 20 refits strongly reduces its objective; larger-scale HWM settings are already used.	not supported
Arbitrary macro vectors are the sole failure	action banks, nearest-code projection, conditional-prior noise, local GMMs, support penalties, and epistemic disagreement do not improve confirmed success.	not primary
Recursive dynamics are too weak	dense supervision reduces high $h = 3$ error $.684 \rightarrow .622$ and low $h = 8$ error $.867 \rightarrow .604$.	real, but insufficient
The high-level score is un-calibrated	exact state-value distillation raises top-1 macro selection $.590 \rightarrow .800$ relative to latent distance.	improved, not solved
The lower controller cannot realize waypoints	measured endpoint residuals and open-loop reachability costs are negative; even true-waypoint control only matches the flat controller in the latest long-tail audit.	remaining gate

These findings motivate two separable tests: learn controller-specific executability from realized lower-controller outcomes, and restrict planning to macro actions obtained by encoding actual text spans. The former tests the high-low interface; the latter changes the optimization domain.

Controller-outcome reranking exposes the missing signal

For each continuous CEM decision, we execute the top 32 proposed subgoals for three closed-loop low-level steps and rerank by the measured endpoint.

Reranking signal	success	interpretation
matched flat / ordinary CEM	.260 / .260	reference
endpoint reachability residual	.260	geometric realizability is insufficient
endpoint latent-goal distance, weight 1 / 10	.280 / .300	modest task alignment
exact realized remaining work	.440	strong candidate-set headroom
learned regression / pairwise / listwise score	.260 / .260 / .290	does not recover oracle gain

The exact reranker changes 46% of decisions, reduces remaining work from 3.46 to 2.90, and reduces selected distractors from .754 to .192. The failure is therefore not lack of useful candidates: learned continuous scoring does not identify task-progressing, controller-realizable candidates reliably.

Support-constrained discrete macro planning

We next plan over encoded sequences of observed intent phrases rather than arbitrary latent vectors:

$$u = q_{\phi}(a_t, \dots, a_{t+h-1}), \quad \hat{z}_{t+h} = F_{\text{hi}}(z_t, u).$$

1. Candidate sequences start with an action feasible at the current state; later elements are drawn from unresolved problem actions. The planner is *not* given their future symbolic feasibility.
2. The high-level model recursively scores the encoded macro sequence, and a learned state-conditioned support score penalizes implausible continuations.
3. The first text action of the selected sequence is executed directly; the sequence is discarded and the model replans from the new observation.

This retains a text-grounded action interface and prevents continuous optimizer exploitation by construction. Crucially, the finite proposal bank must be balanced over its first action: that is the action executed by receding-horizon control. A plain lexicographic cap leaks action ordering.

Candidate-order control invalidates the apparent gain

The initially reported discrete planner capped an exponentially expanding sequence list by taking its first 512 entries. This over-represented early first actions. Three large replications reproduced the implementation, not a model effect.

Stylized protocol	flat	hierarchy	first-feasible control
unbalanced H2, 3×500 mean	.326	.376	not measured
balanced arbitrary sequences, $n = 100$	.310	.110	.100
root-balanced learned-support beam	.310	.150	.100
+ one-step value, weight 1 / 10	.310	.250 / .300	.100

The corrected implementation uses equal proposal multiplicity per current action and reproducibly randomizes future candidates. Every evaluation now reports the ordered-action control. No positive discrete hierarchy result is currently supported; planning-matched feasibility closes the small screen to within one point of flat control but does not work without a one-step value.

Corrected hierarchy fails on first-action credit

All rows use root-balanced candidates on the same 100-problem selection split.

Planner signal	success	sequence validity
flat one-step value	.310	n/a
learned-support beam	.150	learned proposal
planning-matched support + one-step value	.320	learned proposal
oracle-feasible sequences, terminal macro value	.170	1.000
oracle-feasible + one-step value	.330	1.000
prefix-aware macro Q alone, scale .1/1/5	.170/.170/.150	learned proposal
prefix-aware macro Q + one-step value	.330/.330/.320	learned proposal

Oracle-feasible six-action plans complete 4.7 necessary computations on average, yet terminal scoring performs poorly. The planner values eventual progress but executes only the first action and replans, allowing useful work to be deferred repeatedly.

Corrected paired confirmation is negative

The best safeguarded configuration was frozen after the selection screen and evaluated on three fresh 500-problem sets. Candidate banks are root-balanced, future proposals are reproducibly randomized, and action menus are shuffled.

Data / planner seed	flat	hierarchy	difference
71 / 871	.318	.310	-.008
72 / 872	.332	.326	-.006
73 / 873	.300	.296	-.004
mean	.317	.311	-.006

The direction is negative in every pair. Hierarchical prediction is useful as an auxiliary representation experiment, but this controller is not included in the cumulative model recipe.

Prediction improves; abstraction is not established

Dense supervision lowers high-level three-step error $.684 \rightarrow .622$ and low-level eight-step error $.867 \rightarrow .604$. The learned macro-state value has correlation $.860$ with remaining work, and the representations are not collapsed: state/action/macro effective ranks are approximately 239/256, 15/16, and 21/32.

These observations establish predictive structure and task-progress content, not emergent semantic abstraction:

- ▶ recent hierarchy stages freeze the primitive state encoder, so hierarchy cannot reorganize the base representation;
- ▶ the high level predicts the same state space three actions later rather than a separately bottlenecked abstract state;
- ▶ stylized iGSM computations are largely homogeneous and commuting, with few natural option or phase boundaries.

The appropriate abstraction test compares token-, phrase-, sentence-, and paragraph-timescale states and measures whether higher levels discard surface features while retaining more future-relevant reasoning variables.

Faithful iGSM exposed a second ordering leak

The official graph iterates parameters in a near-topological order. Returning feasible actions in that order silently supplies a strong solution heuristic.

Action-menu protocol, $n = 100$ long	flat	hierarchy	first feasible
reference parameter order	.330	.880	1.000
problem-specific shuffled order	.330	.310	.330

The model, prompt, problems, and feasible-action sets are unchanged; only menu order differs. Every method now receives the same stable, problem-specific shuffled menu and every table reports the first-feasible control. Earlier faithful planning results using reference order are excluded and must be rerun.

The earlier macro-depth curve is not valid evidence

Depth is selected on a separate 100-problem validation screen using the same checkpoint and candidate budget.

Controller	flat	H1	H2	H3	H4
success	.310	.180	.410	.390	.330

All four depths used the unbalanced sequence cap. Because truncation severity changes with sequence length, the curve conflates planning depth with which first actions survive enumeration. It is retained only as provenance and is excluded from model selection. Depth must be rerun after the corrected, root-balanced proposal mechanism is fixed.

What the evidence currently supports

1. Intent-conditioned latent prediction supports auditable counterfactual transitions and closed-loop reasoning.
2. Action selection, not one-step transition identity, is the main source of residual episode failures.
3. Raw-action displacement decoding prevents deterministic collapse and strengthens variational transition grounding, but does not by itself ensure healthy probabilistic rank.
4. Dense low- and high-level supervision improves recursive prediction; learned next-state values are better calibrated than latent goal distance. Continuous latent-action CEM nevertheless fails paired confirmation.
5. Restricting macro actions to encoded text spans does not improve control: the initial positive result was caused by first-action-biased candidate truncation, and corrected 3×500 planning is .311 versus .317 flat.
6. Inferred posterior transition codes should not be presented as usable actions until a state-conditioned prior can retrieve or generate them. Hard-text hierarchy remains paused. Faithful iGSM also requires a shuffled action menu: the reference parameter iteration order is itself a topological-order leak.

Limitations and completion criteria

- ▶ The flat stylized model and policy baselines have three validation seeds. The corrected hierarchical planner failed paired confirmation; the final task test remains sealed until the flat cumulative and ablation matrix is frozen.
- ▶ The latent-goal teacher uses environment interactions and a demonstrated terminal state. It removes symbolic quality labels, but not the action interface.
- ▶ Flat lookahead beyond one step requires the reference graph to propose future feasible actions and is therefore an oracle-action diagnostic. The deployed discrete hierarchy uses current feasible actions only for the first choice; future candidates are unresolved text actions, not oracle-feasible sequences.
- ▶ Exact remaining graph distance substantially improves the high-level state value. It is an oracle-supervised mechanism diagnostic until a non-symbolic target reproduces the effect.
- ▶ HWM-style terminal latent distance requires an encoded solved-state goal observation. That observation is available in goal-conditioned control but not in answer-unknown reasoning, so it is an oracle diagnostic here.
- ▶ Official iGSM requires its own objective-selection and ablation cycle; stylized choices are hypotheses, not guaranteed transfers.
- ▶ Probabilistic action-free models currently lack a usable action prior. They are a separate representation study and are paused during hierarchy work.

Root-balanced planning did not exceed matched flat control on three independent 500-problem sets, so hierarchy is excluded from the easy-domain main recipe. Any future faithful transfer must randomize action-menu order and use a newly predeclared protocol rather than inheriting the invalid historical result.

Appendix: artifact mapping

Scientific condition	Artifact family
multi-step latent-goal preference distillation	disc_gar_greedy_*
raw-action displacement stability factorial	dldad_*
minimal text Delta-JEPA controls	deltajepa_text_*
probabilistic observed-action models	dvjepa_*
action-free sentence stream	svjepa_*
fair intent-policy language models	lm_intent, sentlm_*_intent
official-iGSM transfer	real_*, faithful configuration suffixes
dense high-level recursive supervision	intent_hidense_tail79_n*
dense low-level recursive supervision	intent_lodense_tail79_n*
macro-state value calibration	intent_hivalue_tail79_*
paired hierarchy confirmation	confirm_ds*_s*_{flat,hier}.json
controller-outcome diagnostics	intent_controller_outcomes/*
discrete hierarchy confirmation	confirm_ds*_s*_{h2,discrete_h3}.json

Appendix: diagnostics reported for every final model

- ▶ optimal-budget and +2-action closed-loop success;
- ▶ chosen-action distractor rate;
- ▶ counterfactual transition nearest-neighbour matching and RSA;
- ▶ value-energy pairwise τ and oracle-order top-1;
- ▶ state scale and effective rank;
- ▶ observed-action token and exact-phrase recovery when LDAD is present;
- ▶ probabilistic NLL, standardized residual, and one/two-sigma coverage;
- ▶ posterior/prior semantic probes and retrieval for inferred actions.